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MTH:2310 DISCRETE MATH
QUIZ 6
DUE: FRIDAY OCT. 24

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. *(14 points) Solving Linear Recurrence Relations*
Suppose $(r-2)^3(2r-3)^2(3r-1)(r+1)(r+2)^2 = 0$ is the characteristic equation for a recurrence relation.

(a) *(2 points)* What is the degree of the recurrence relation?

(b) *(4 points)* Find all characteristic roots and state their multiplicity.

(c) *(8 points)* What is the general form of the solution to the recurrence relation?

Problem 2. *(12 points) Solving Linear Recurrence Relations*
Consider the recurrence relation $a_n = -9a_{n-1} - 27a_{n-2} - 27a_{n-3}$ with initial conditions $a_0 = -4$, $a_1 = 6$, $a_2 = 18$.

(a) *(3 points)* Find the characteristic roots associated to the recurrence relation. State their multiplicities.

(b) *(4 points)* State the general form of the solution to the recurrence relation.

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(c) *(5 points)* Find the exact solution to the recurrence relation.

Problem 3. *(12 points) Mathematical Induction*

Prove that for every positive integer n, $\dfrac{1}{1\cdot 2}+\dfrac{1}{2\cdot 3}+\cdots+\dfrac{1}{n\cdot(n+1)} = \dfrac{n}{n+1}$.

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Problem 4. *(12 points) Mathematical Induction*
Prove that for any non-negative integer n, 6 divides $7^n - 1$. Recall that a non-zero integer b divides an integer a, if there is an integer q such that $b = qa$.